MTE 203 – Advanced Calculus
Week 4 Tutorial

Problem 1: Midterm Exam Problem 4 Fall 2016

A bead of mass "m" slides without friction down a wire bent in the shape of the curve $y = x^2$ under the influence of the gravitational force $-mg\hat{\jmath}$.

The speed of the bead as it passes through the point $(1,1)$ is "V". At that instant, find

a. The normal acceleration of the bead in terms of "V"

b. The rate of change of its speed in terms of "g".

Problem 2: [S.12.1, Prob. 27]

a. A company wishes to construct a storage tank in the form of a rectangular box. If material for sides and top costs \$1.25/m^2 and material for the bottom costs \$4.75/m^2, find the cost of building the tank as a function of its length l, width w and height h.

b. If the tank must hold 1000 m^3, find the construction cost in terms of l and w.

Problem 3 [S.12.1, Prob. 17]

Draw the level curves of $f(x, y) = y - x^2$ in a contour map corresponding to the values $f(x, y) = -2, -1, 0, 1, 2$